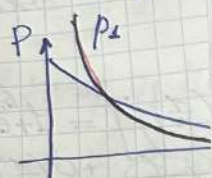


$$\sup_{\theta_0} L(\vec{x}_n) = L_0(\vec{x}_n) = \theta_0 e^{-\theta_0 x}$$

$$\sup_{\theta > \theta_0} \theta e^{-\theta x}$$

$$\frac{\partial}{\partial \theta} \theta e^{-\theta x} = -x e^{-\theta x}$$

$$l(\vec{x}_n) = \frac{1}{x} \ln \theta_0 e^{-\theta_0 x}$$



T 13

$$n_1 = 139$$

$$n_2 = 1000$$

$$S_{1L} = 5.722 = a_1$$

$$S_{1L} = 6.161 = b_1$$

$$S_{1W} = 4.612 = a_2$$

$$S_{1W} = 5.055 = b_2$$

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$H_1: \sigma_1^2 \neq \sigma_2^2$$

$$\frac{(n_1 - 1) S_x^2}{a_1^2} \sim \chi^2(n_1 - 1)$$

$$\frac{(n_2 - 1) S_y^2}{b_1^2} \sim \chi^2(n_2 - 1)$$

$$\Delta = \frac{S_x^2 / a_1^2}{S_y^2 / b_1^2} \sim F(n_1 - 1, n_2 - 1)$$

$$P(\Delta \leq C_1 | H_0) = \frac{\alpha}{2}$$

$$C_1 = 0.767$$

$$P(\Delta > C_2 | H_0) = \frac{\alpha}{2}$$

$$C_2 = 1.303$$

$$G: \tilde{\Delta} \leq F_{\frac{\alpha}{2}}(n_1 - 1, n_2 - 1)$$

$$\tilde{\Delta} \geq F_{\frac{\alpha}{2}}(n_2 - 1, n_1 - 1)$$

$$\tilde{\Delta} = \frac{S_x^2}{S_y^2} =$$

$$P\text{-value} = 2 P(\Delta \geq \tilde{\Delta})$$

$$= \left(\frac{4.612}{5.055} \right)^2 = 0.832$$

$$0.767 < 0.832 < 1.303 \quad \checkmark$$

Нем оснований отвергнуть H_0 при уровне

$$\tilde{\Delta} = \left(\frac{5.722}{6.161} \right)^2 = 0.862$$

$$0.767 < 0.862 < 1.303 \quad \checkmark$$

Нем оснований отвергнуть H_0 при уровне

$$W = P(|\Delta|)$$

$$W = P(C_1 \leq \Delta \leq C_2 | H_0) =$$

$$= P\left(\frac{S_x^2}{S_y^2} < \frac{C_1^2}{C_2^2} C_1 | H_0\right) + P\left(\frac{S_x^2}{S_y^2} > C_2 \frac{C_1^2}{C_2^2} | H_0\right)$$

$$T14 \quad H_0: a=b \quad H_1: a>b$$

$$\bar{x} = \frac{-1,11 - 6,10 + 2,42}{3} = -1,596$$

$$\bar{y} = -2,6$$

$$\frac{(\bar{x} - \bar{y}) - (a - b)}{\sqrt{\frac{\tilde{\sigma}_x^2}{n} + \frac{\tilde{\sigma}_y^2}{m}}} \stackrel{\text{при } H_0}{=} 0,929$$

$$\sim N(0,1)$$

$$U_{1-\alpha} = 1,6449$$

$$0,929 < 1,645 \text{ — не отвергаем } H_0$$

$$p\text{-value} = P\left(\frac{\bar{x} - \bar{y}}{\sqrt{\frac{\tilde{\sigma}_x^2}{n} + \frac{\tilde{\sigma}_y^2}{m}}} > 0,929 | H_0\right) =$$

$$= 1 - \underbrace{\Phi(0,93)}_{\substack{\text{крит.} \\ \text{знач.} \\ 0,82}} = 0,48 > \alpha \text{ не отвергаем } H_0$$

$$W = P\left(\frac{\bar{x} - \bar{y} - (a - b)}{\sqrt{\frac{\tilde{\sigma}_x^2}{n} + \frac{\tilde{\sigma}_y^2}{m}}} > 1,645 | H_1\right) =$$

$$= 0,237$$

$$W = P$$

$$W(\Delta) = 1 - \Phi\left(1,645 - \frac{\Delta}{\sqrt{\frac{\sigma_x^2}{n} + \frac{\sigma_y^2}{m}}}\right)$$

$$W(\Delta) = 1 - \Phi(1,645 - 0,926 \cdot \Delta)$$

